

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Eighth Semester B.E. (E&C) (Credit System) Degree Examinations

May- June 2023

19ECE71 – BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** and **Unit – II** each and **One full** question from **Unit – III**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Define Big Data. What are the four types of Data structures in Big Data? Explain with an example for each. | 10 | L*2 | 1 | 1 |
| b) Briefly describe the Analyst Perspective on Data Repositories. What are the key skill sets and behavioral characteristics of a data scientist? | 10 | L2 | 1 | 1 |
| 2. a) Write a R program
i. To print Fibonacci series.
ii. To make a simple calculator using functions. | 10 | L3 | 2 | 2 |
| b) Mention the different stages of Data analytics life cycle. Briefly explain the responsibilities of any four stages. | 10 | L2 | 1 | 1 |
| 3. a) Explain any 5 types of R-objects with the help of programming instruction and result for each. | 10 | L2 | 1 | 1 |
| b) Write a short note on
I. Hypothesis testing
II. Type I and Type II errors | 10 | L2 | 2 | 1 |

Unit – II

- | | | | | |
|--|----|----|---|---|
| 4. a) Explain evaluation of candidate rules. Give an example for each. Mention the application of association rules considering market basket analysis. | 10 | L2 | 4 | 1 |
| b) What are decision trees in classification. Give an overview of decision tree with the help of an example. Write the general algorithm of decision tree. | 10 | L2 | 3 | 1 |
| 5. a) Discuss the components included in Time Series analysis. Explain Box-Jenkins Methodology in Time Series analysis. | 10 | L2 | 4 | 1 |
| b) | | | | |

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Calculate the likelihood of playing under the specified conditions:

Outlook = Sunny

Temperature= cool

Humidity = high

Windy = true

Using Bayes theorem find out the probability of play (yes and no)

6. a) Consider the following Confusion Matrix

Actual		Predicted	
		Subscribe	Not Subscribed
	Subscribe	3	8
	Not Subscribed	2	87

Find:

- i. Accuracy.
- ii. True Positive Rate.
- iii. False Positive Rate.
- iv. False Negative Rate.
- v. Precision.

10 L3 3 2

- b) What is clustering and describe the overview of the method of k-means clustering in data analytics using four steps. Divide the given sample data into 2 clusters using K means clustering algorithm {2,5,7,8,13,16,18,28,30}

10 L3 4 2

Unit – III

7. a) Write short notes on NoSQL and MADlib functions
b) With a neat diagram explain architecture of Hive

10 L2 5 1

10 L2 5 1

8. a) What is HDFS? Discuss the features of HDFS and comment why do we need this system.
b) With a help of examples explain DDL, DCL, DML, TCL commands.

10 L2 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
